



Lab Sheet 2

Data Structures– I Lab(BTCS209P)

B. Tech.(AIML,CyberSecurity,andDataScience)

III Semester (Session 2026–2027)

Objective

To understand and implement fundamental array operations such as traversal, insertion, deletion, searching, and sorting in C++.

Prerequisites

- Basic knowledge of C++ programming and loops.
- Understanding of arrays and array indexing.
- A C++ compiler (Code::Blocks / Visual Studio Code / Dev-C++).

Task

Open your C++ IDE and perform the following tasks.

Task 1: Array Traversal

Read N elements into an array and display the elements, indices, sum, and average.

Task 2: Array Insertion

Write a C++ program to insert an element at the beginning, end, and a given index. Display the array before and after insertion.

Task 3: Array Deletion

Write a C++ program to delete an element from the beginning, end, and a given index. Display the array before and after deletion.

Task 4: Searching and Sorting

Search an element using Linear Search and sort the array using Bubble Sort in ascending order.

Task 5: Menu-Driven Array Operations

Create a menu-driven program to display the array, insert, delete, search, sort, and exit. Test valid and invalid inputs.

Instructions

- Use 0-based indexing for array operations.
- Use meaningful variable names, proper indentation, and comments.
- Handle invalid indices and boundary conditions appropriately.
- Compile and execute each program successfully.
- Save each program with an appropriate filename.

Submission

- Take screenshots of source code and output for each task.
- Save all C++ source files in a single folder.
- Submit the folder along with the screenshots to the course instructor.

LAB SHEET--2

Task

Open your C++ IDE and perform the following tasks.

Task 1: Array Traversal

Read N elements into an array and display the elements, indices, sum, and average.

solution :

```
#include <iostream>
using namespace std;
int main()
{
    int Arr[20],n,i,sum=0;
    cout<<"Enter the number of element of an array :";
    cin>>n;
    cout<<"Enter the elements of an array :";
    for(int i=0;i<n;i++){
        cin>>Arr[i];
        sum+=Arr[i];
    }
    cout<<"Indices of an array element are :";
    for(int i=0;i<n;i++){
        cout<<"Index"<<" "<<i<<"="<<Arr[i]<<endl;
    }
    cout<<"Sum of an array element:"<<sum<<endl;
    int avg = sum/n;
    cout<<"Average of an array element :"<<avg<<endl;
    return 0;
}
```

Task 2: Array Insertion

Write a C++ program to insert an element at the beginning, end, and a given index. Display the array before and after insertion.

solution :

```
#include <iostream>
using namespace std;

int main()
{
    int arr[100];
    int n, choice, value, index;

    cout << "Enter number of elements: ";
    cin >> n;

    if (n < 0 || n > 99)
    {
        cout << "Invalid size!" << endl;
        return 0;
    }
    cout << "Enter array elements:" << endl;
    for (int i = 0; i < n; i++)
    {

    }

    cin >> arr[i];
```

```

}do
{
    cout << endl << "===== ARRAY INSERTION
===== " << endl;
    cout << "1. Insert at Beginning" << endl;
    cout << "2. Insert at End" << endl;
    cout << "3. Insert at Given Index" << endl;
    cout << "4. Display Array" << endl;
    cout << "5. Exit" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice)
    {
    case 1:
        if (n >= 100)
        {
            cout << "Array is full!" << endl;
            break;
        }
        cout << "Enter element: ";
        cin >> value;
        cout << "Array before insertion:" << endl;
        (int i = 0; i < n; i++)
        cout << arr[i] << " ";
        for (int i = n; i > 0; i--)
        {
            arr[i] = arr[i - 1];
        }
        arr[0] = value;
        n++;
        cout << endl << "Array after insertion:" << endl;
        for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
        cout << endl;
        break;

```

case 2:

```
if (n >= 100)
{
    cout << "Array is full!" << endl;
    break;
}
cout << "Enter element: ";
cin >> value;
cout << "Array before insertion:" << endl;
for (int i = 0; i < n; i++)
    cout << arr[i] << " ";
arr[n] = value;
n++;
cout << endl << "Array after insertion:" << endl;
for (int i = 0; i < n; i++)
    cout << arr[i] << " ";
cout << endl;
break;
```

case 3:

```
if (n >= 100)
{
    cout << "Array is full!" << endl;
    break;
}
cout << "Enter index (0 to " << n << "): ";
cin >> index;
if (index < 0 || index > n)
{
    cout << "Invalid index!" << endl;
    break;
}
```

```

cout << "Enter element: ";
    cin >> value;
    cout << "Array before insertion:" << endl;
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    for (int i = n; i > index; i--)
    {
        arr[i] = arr[i - 1];
    }
    arr[index] = value;
    n++;
    cout << endl << "Array after insertion:" << endl;
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
    break;
case 4:
    cout << "Current Array:" << endl;
    for (int i = 0; i < n; i++){
        cout << "Index " << i << " = " << arr[i] << endl;
    }
    break;
case 5:
    cout << "Exiting..." << endl;
    break;
default:
    cout << "Invalid choice! Please try again." << endl;
}
} while (choice != 5);
return 0;
}

```

Task 3: Array Deletion

Write a C++ program to delete an element from the beginning, end, and a given index. Display the array before and after deletion.

solution

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int arr[100];
    int n, choice, index;

    cout<<"Enter number of elements: ";
    cin >> n;

    if (n <= 0 || n > 100)
    {
        cout<<"Invalid size!"<< endl;
        return 0;
    }

    cout<<"Enter array elements:" << endl;

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }
```

```

do
{
    cout << endl << "===== ARRAY DELETION ====="
    << endl;

    cout << "1. Delete from Beginning" << endl;
    cout << "2. Delete from End" << endl;
    cout << "3. Delete from Given Index" << endl;
    cout << "4. Display Array" << endl;
    cout << "5. Exit" << endl;
    cout << "Enter your choice: ";
    cin >> choice;
    switch (choice)
    {
    case 1:
        if (n == 0)
        {
            cout << "Array is empty!" << endl;
            break;}
        cout << "Array before deletion:" << endl;
        for (int i = 0; i < n; i++)
            cout << arr[i] << " ";
        for (int i = 0; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        n--;

```

```

cout<<endl<< "Array after deletion:" << endl;
    for(int i=0; i < n; i++)
        cout<<arr[i] << " ";
    cout << endl;
    break;
case 2:
    if (n == 0) {
        cout<<"Array is empty!" << endl;
        break;
    }
    cout<<"Array before deletion:" << endl;
    for(int i=0; i < n; i++)
        cout<<arr[i] << " ";
    n--;
    cout<<endl << "Array after deletion:" << endl;
    for(int i=0; i < n; i++)
        cout<<arr[i] << " ";
    cout << endl;
    break;
case 3:
    if (n == 0){
        cout<<"Array is empty!" << endl;
        break;
    }

```

```

cout << "Enter index (0 to " << n - 1 << "): ";
cin >> index;

if (index < 0 || index >= n)
{
cout << "Invalid index!" << endl;
break;
}
cout<<"Arraybeforedeletion:" << endl;

    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    for(inti=index;i<n - 1; i++) {
        arr[i] = arr[i + 1];
    }
    n--;
    cout<<endl<<"Array after deletion:" << endl;
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
    break;
case 4:
    if (n == 0) {
        cout<<"Arrayisempty!" << endl;
    }

```

```
c else
{
cout << "Current Array:" << endl;
for (int i = 0; i < n; i++)
{
cout << "Index " << i << " = " << arr[i] << endl;
}
}
break;
case 5:
cout << "Exiting..." << endl;
break;
default:
cout << "Invalid choice! Please try again." << endl;
}
} while (choice != 5);
return 0;
}
```

Task 4: Searching and Sorting

Search an element using Linear Search and sort the array using Bubble Sort in ascending order.

solution :

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int arr[100], n;
```

```
    cout<<"Enter number of elements: ";
```

```
    cin >> n;
```

```
    if(n<=0||n>100){
```

```
        cout<<"Invalid size!"<< endl;
```

```
        return 0;
```

```
    }
```

```
    cout<<"Enter array elements:" << endl;
```

```
    for (int i = 0; i < n; i++){
```

```
        cin >> arr[i];
```

```
    }
```

```
    for(int i=0; i<n-1; i++) {
```

```
        for(int j=0; j<n-i-1; j++) {
```

```
            if (arr[j] > arr[j + 1]){
```

```
                int temp = arr[j];
```

```
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
cout << "Array after Bubble Sort:" << endl;
for (int i = 0; i < n; i++){
    cout << arr[i] << " ";
}
cout << endl;
return 0;
}
```

Task 5: Menu-Driven Array Operations

Create a menu-driven program to display the array, insert, delete, search, sort, and exit. Test valid and invalid inputs.

solution :

```
#include <iostream>
using namespace std;
int main()
{
    int arr[100];
    int n, choice;
    int value, index;
    bool found;
    cout<<"Enter number of elements: ";
    cin >> n;
    if (n < 0 || n > 100){
        cout<<"Invalid size!" << endl;
        return 0;
    }
    cout<<"Enter array elements:" << endl;
    for (int i = 0; i < n; i++){
        cin >> arr[i];
    }
    do{
        cout<<endl<<"===== " << endl;
        cout<<"  ARRAY OPERATIONS " << endl;
        cout<<"===== " << endl;
```

```
cout << "1. Display Array" << endl;
cout << "2. Insert Element" << endl;
cout << "3. Delete Element" << endl;
cout << "4. Search Element" << endl;
cout << "5. Sort Array" << endl;
cout << "6. Exit" << endl;
cout << "Enter your choice: ";
cin >> choice;
switch (choice){

    case 1:
        if (n == 0) {
            cout << "Array is empty!" << endl;
        }
        else
        { cout << "Array elements:" << endl;
          for (int i = 0; i < n; i++)
              { cout << "Index " << i << " = " << arr[i] << endl;
                }
          }
        break;
```

```
case 2:
if (n >= 100)
{
cout << "Array is full!" << endl;
break;
}
cout << endl << "1. Beginning" << endl;
cout << "2. End" << endl;
cout << "3. Given Index" << endl;
cout << "Enter insertion choice: ";
int insertChoice;
cin >> insertChoice;
cout << "Enter element: ";
cin >> value;
if (insertChoice == 1) {
for (int i = n; i > 0; i--){
arr[i] = arr[i - 1];
}
arr[0] = value;
n++;
cout << "Element inserted at beginning." << endl;
}
```

```

else if (insertChoice == 2) {
    arr[n] = value;
    n++;
    cout << "Element inserted at end." << endl; }
else if (insertChoice == 3){
    cout << "Enter index (0 to " << n << "): ";
    cin >> index;
    if (index < 0 || index > n) {
        cout << "Invalid index!" << endl; }
    else
    { for (int i = n; i > index; i--)
        {
            arr[i] = arr[i - 1];
        }
        arr[index] = value;
        n++;
        cout << "Element inserted successfully." << endl;
    }
}
else
{cout << "Invalid insertion choice!" << endl;
}
break;

```

case 3:

```
if (n == 0){
    cout<<"Arrayisempty!"<< endl;
    break;
}
cout<<endl<<"1.Beginning" << endl;
cout << "2. End" << endl;
cout << "3. Given Index" << endl;
cout<<"Enterdeletionchoice: ";
int deleteChoice;
cin >> deleteChoice;
if (deleteChoice == 1) {
    for(inti=0;i<n-1;i++){
        arr[i] = arr[i + 1];
    }
    n--;
    cout<<"Elementdeletedfrom beginning." << endl;
}
else if (deleteChoice == 2) {
    n--;
    cout<<"Elementdeletedfrom end." << endl;
}
```

```

else if (deleteChoice == 3) {
cout << "Enter index (0 to " << n - 1 << "): ";
cin >> index;
if (index < 0 || index >= n)
{cout << "Invalid index!" << endl;
}
else
{for (int i = index; i < n - 1; i++)
{
arr[i] = arr[i + 1];
}
n--;
cout << "Element deleted successfully." << endl;
}
}
else
{
cout << "Invalid deletion choice!" << endl;
}
break;
case 4:

    cout << "Enter element to search: ";
    cin >> value;
    found = false; JAYSHREESAINI(CYBERSECURITY) ,( CU25230014),E

```

```
for (int i = 0; i < n; i++) {  
    if (arr[i] == value){  
        cout << "Element found at index " << i << endl;  
        found = true;  
        break;  
    }  
}  
  
if (!found){  
    cout << "Element not found!" << endl;  
}  
  
break;
```

case 5:

// Bubble Sort

```
    for (int i = 0; i < n - 1; i++){  
        for (int j = 0; j < n - i - 1; j++){  
            if (arr[j] > arr[j + 1]) {  
                int temp = arr[j];  
                arr[j] = arr[j + 1];  
                arr[j + 1] = temp;  
            }  
        }  
    }  
  
    cout << "Array sorted using Bubble Sort:" << endl;
```

```
for (int i = 0; i < n; i++){  
    cout << arr[i] << " ";  
}  
cout << endl;  
break;  
case 6:  
    cout << "Program exited successfully." << endl;  
break;  
default:  
    cout << "Invalid choice! Please enter 1 to 6." << endl;  
}  
} while (choice != 6);  
return 0;  
}
```